2.4 Classwork: Constant Acceleration Practice Solutions

1. Vernier cart. Knowns: $v_0 = 0$, $a = 0.200$ m/s^2, $x_0 = 0$.

   (a)

t (s)	0	1.00	2.00	3.00	4.00
v (m/s)	**0**	**0.200**	**0.400**	**0.600**	**0.800**

   Each second adds 0.200 m/s; this is $v = v_0 + at$ one row at a time.

   (b) Want $x - x_0$ at $t = 3.00$ s. Use $x = x_0 + v_0 t + \frac{1}{2}at^2$:
   $x - x_0 = 0 + \frac{1}{2}(0.200 \text{ m/s}^2)(3.00 \text{ s})^2 = \mathbf{0.900\ m}$
   Also accept the average velocity route, using the table: $\bar{v} = \frac{0.600+0}{2} = 0.300$ m/s, and 0.300 m/s $\times$ 3.00 s $= 0.900$ m. A common slip is $\frac{1}{2}(0.200)(3.00) = 0.300$ m, forgetting to square the time.

2. Taipei 101 elevator. Knowns: $v_0 = 0$, $v = 16.8$ m/s, $a = 1.20$ m/s^2.

   (a) Want t. Use $v = v_0 + at$, so $t = \dfrac{v - v_0}{a}$:
   $$t = \frac{16.8 \text{ m/s} - 0}{1.20 \text{ m/s}^2} = \mathbf{14.0\ s}$$

   (b) Use $\bar{v} = \dfrac{v + v_0}{2}$:
   $$\bar{v} = \frac{16.8 \text{ m/s} + 0}{2} = \mathbf{8.40\ m/s}, \text{ upward.}$$

   (c) Want $x - x_0$. Use $x - x_0 = \bar{v}\, t$:
   $x - x_0 = 8.40$ m/s $\times$ 14.0 s $= 117.6 \approx \mathbf{118\ m}$
   Also accept $v^2 = v_0^2 + 2a(x - x_0)$: $x - x_0 = \frac{(16.8)^2}{2(1.20)} = \frac{282.24}{2.40} = 117.6$ m, or $\frac{1}{2}at^2 = \frac{1}{2}(1.20)(14.0)^2 = 117.6$ m.

3. Driver at 30 mi/h. Take the direction of travel as positive.

   (a) Knowns: $v_0 = 13.4$ m/s, $a = 0$, $x - x_0 = 9$ m. Want t. With $a = 0$, $x = x_0 + v_0 t + \frac{1}{2}at^2$ becomes $x - x_0 = v_0 t$, so

   $$t = \frac{x - x_0}{v_0} = \frac{9 \text{ m}}{13.4 \text{ m/s}} = 0.6716\ldots \approx \mathbf{0.67 \text{ s}}$$

   Accept 0.7 s: 9 m has one significant figure, so 0.7 s is the strict answer. This is longer than the 0.20–0.25 s reaction time of a catch-the-ruler test, because a driver has to notice the hazard, decide, and move a foot to the brake.

   (b) Knowns: $v_0 = 13.4$ m/s, $v = 0$, $a = -6.4$ m/s^2. Want t. Use $v = v_0 + at$, so $t = \dfrac{v - v_0}{a}$:

   $$t = \frac{0 - 13.4 \text{ m/s}}{-6.4 \text{ m/s}^2} = 2.09375 \approx \mathbf{2.1 \text{ s}}$$

   Two significant figures, because -6.4 m/s^2 has two. Accept 2.09 s. Check: $\bar{v} = \frac{13.4+0}{2} = 6.7$ m/s, and 6.7 m/s $\times$ 2.09375 s $= 14.03$ m, the Highway Code's 14 m.

   (c) Total time: $0.6716\ldots$ s $+ 2.09375$ s $= 2.765\ldots \approx \mathbf{2.8 \text{ s}}$

   Total distance: 9 m $+ 14$ m $= \mathbf{23 \text{ m}}$

   Add the unrounded times. 23 m is the overall stopping distance the Highway Code prints for 30 mi/h; about two-fifths of it is covered before the brakes are touched.

4. Frecciarossa 1000. Knowns: $v_0 = 83.3$ m/s, $v = 0$, $a = -0.500$ m/s^2.

   (a) Want t. Use $v = v_0 + at$, so $t = \dfrac{v - v_0}{a}$:

   $$t = \frac{0 - 83.3 \text{ m/s}}{-0.500 \text{ m/s}^2} = 166.6 \approx \mathbf{167 \text{ s}}$$

   About 2.8 minutes. The two negatives cancel, so the time comes out positive, as it must.

   (b) Want $x - x_0$. Use $\bar{v} = \dfrac{v + v_0}{2} = \dfrac{0 + 83.3}{2} = 41.65$ m/s:

   $$x - x_0 = \bar{v}\, t = 41.65 \text{ m/s} \times 166.6 \text{ s} = 6938.89 \approx 6940 \text{ m} \approx \mathbf{6.94 \text{ km}}$$

   Also accept $v^2 = v_0^2 + 2a(x - x_0)$: $x - x_0 = \frac{0-(83.3)^2}{2(-0.500)} = \frac{-6938.89}{-1.000} = 6938.89$ m. Carry 166.6 s, not the rounded 167 s, into this step; 167 s gives 6955.6 m $\approx$ 6.96 km. The driver starts braking nearly 7 km out, far beyond where the station can be seen.